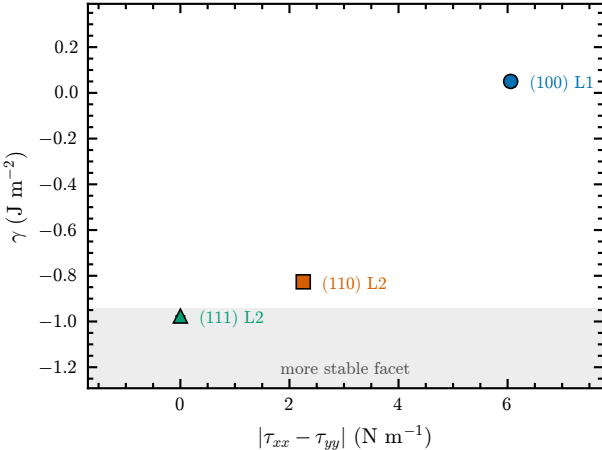




Surface energy against the magnitude of the surface-stress anisotropy, one point per facet. γ is read from surface_energy.csv, computed as $[E^{\text{slab}} - N^{\text{C}} \cdot \mu^{\text{C}} - N^{\text{H}} \cdot \mu^{\text{H}}] / 2A$ at the H-rich limit where μ^{H} is half the energy of a relaxed H₂ molecule at the same 90/720 Ry setting; error bars are the scatter of γ across the fitted thickness ladder. Negative γ means the H-terminated face is more stable than bulk diamond plus H₂, which is expected because hydrogenation is exothermic. The two quantities are anticorrelated: (111) is the most stable facet and carries exactly zero anisotropy, while (100) carries all of it and is not measurably stabilised by H termination at all ($+0.050 \pm 0.024$ J/m², i.e. indistinguishable from zero within its own ladder scatter). Engineering a transverse NV splitting therefore means favouring the facet that equilibrium morphology suppresses, i.e. a non-equilibrium particle shape. Epistemic level differs by surface ((111) L2; (110) L2; (100) L1): the (100) ladder does not yet support a converged γ , so its position on this axis is provisional while its anisotropy is not. γ shifts with the H chemical potential at up to 3.5 J/m² per eV, so these are H-rich values and not the whole phase diagram. The shaded direction marks lower γ , not a Wulff region: a Wulff construction is undefined when γ is negative.